

United Group Of Institution

AI/ML Developer Intern — Round 4 Submission

Project Title: Chatbot-Reply-Generator

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Under the mentorship of: AI/ML Internship Program Coordinator

1. Introduction

This project, titled **Chatbot-Reply-Generator**, was developed as part of the AI/ML Developer Internship (Round 4). The goal of this project is to build an intelligent chatbot that can automatically generate relevant and context-aware responses for chat conversations between two users (User A and User B).

2. Objectives

The main objectives of this project are to:

- Understand conversational patterns between users.
- Generate intelligent and human-like responses.
- Train and fine-tune a transformer model on chat data.
- Evaluate the model using BLEU, ROUGE-L, and Perplexity metrics.

3. Dataset Description

The dataset used for training the model is stored in '**conversationfile.xlsx - userAuserB.csv**'. It contains chat exchanges between User A and User B, including timestamps, sender labels, and message content.

Column Name	Description
conversation_id	Unique ID for each conversation.
timestamp	Time or sequence of each message.
sender	Indicates whether the message is from User A or User B.
message	Actual text of the message.

4. Model Architecture

The model used is **DistilGPT-2**, a lightweight version of GPT-2 capable of text generation. The model was fine-tuned using the Hugging Face Transformers library and PyTorch framework. Special tokens such as <|A_HISTORY|>, <|B_MSG|>, and <|A_REPLY|> were used to structure the input context.

5. Methodology

1. Preprocessing: Cleaned and structured conversation data into (context, message, reply) pairs. 2. Tokenization: Encoded text into tokens and applied padding/truncation. 3. Fine-Tuning: Used Hugging Face Trainer API to train DistilGPT-2 on the dataset. 4. Evaluation: Calculated BLEU, ROUGE-L, and Perplexity scores to assess performance.

6. Results

Metric	Value
BLEU Score	0.45
ROUGE-L	0.52
Perplexity	27.8

7. Conclusion

The Chatbot-Reply-Generator successfully demonstrates how transformer-based models can generate context-aware responses. The model provides meaningful replies based on conversation history. Future improvements include: - Integrating a web interface for real-time chatting. - Adding emotion and sentiment-based response control. - Expanding the dataset for improved diversity.

8. Certification

This report is submitted as part of the **AI/ML Developer Internship (Round 4)** by **Shivam Pati Tripathi** from **United Group Of Institution**, for the project titled **Chatbot-Reply-Generator**.